The Misspecification Frontier: When Flexible-Shape Quantile Models Beat the GARCH Industry Standard, and a Deployable Engine That Captures the Edge*

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Draft — September 4, 2026

Abstract

We propose a deployable downside-risk engine and a theory of when it pays. The engine — *amortized residual-hybrid* forecasting — keeps the GARCH conditional scale that industry models get right and replaces only the fixed innovation shape with a state-conditioned nonparametric quantile model trained *once, across hundreds of names*, finished with an extreme-value tail and a conformal recalibration. Three results. *First, amortization:* one fit replaces per-asset estimation, transfers to names never seen in training, beats own-history benchmarks at every listing age (6–10% of pinball loss for young listings), and prices assets with *no return history* — day-one risk numbers for IPOs that no per-asset model can produce. *Second, the engine beats what banks run:* against historical simulation, RiskMetrics, GARCH-t, GJR-skew-t, and filtered historical simulation on 140 CRSP names, it wins on average loss with date-level Diebold–Mariano significance (statistics 4.4–8.3), delivers the best-calibrated 97.5% Expected Shortfall — removing an 8–11% ES over-statement that over-charges capital — and with its tail stages its exception counts are consistent with nominal at both regulatory levels (pooled panel diagnostic; per-asset restatement registered); the edge grows at the ten-day horizon and holds in the 2020–22 stress window. *Third, a misspecification frontier says when the edge is large:* a real-time score — the excess kurtosis and asymmetry of recent GARCH-standardized residuals — concentrates the gain (+2.7%, DM 6.5, in its top decile; 12–14% on hyperinflation currencies) and maps it across US equities, FX, twenty-six country indices, and forty-three cross-asset instruments. Where the score is quiet the engine ties the parametric benchmark, so the deployment rule is one line: run the engine everywhere, read the score as a risk dial.

*Paper A of a three-paper program. The neural implicit-quantile-network estimator, the EVT tail graft, and the cross-sectional conformal layer used as one stage of the engine below are developed in the companion paper *GRAFT-Q* (Qin, 2026); here the deployable estimator is gradient-boosted trees and those components are invoked by reference. The generative-posterior, simulation-based-calibration, and multivariate results are developed in the companion paper (Paper B). All results are out-of-sample under the walk-forward protocol described in the text; code and data availability are stated at the end.

Keywords: Value-at-Risk; Expected Shortfall; FRTB; quantile regression; extreme value theory; conformal prediction; misspecification; amortized estimation.

JEL classification: C14; C22; C52; C58; G17; G28.

1 Introduction

Two communities measure conditional downside risk. Econometrics builds parametric filters — GARCH and its leverage and skew extensions — whose conditional scale dynamics are excellent and whose innovation law is fixed by assumption.[1] Machine learning builds distribution-free conditional quantile estimators — gradient-boosted quantile trees, neural implicit quantile networks (IQN), and, most recently, generative Bayesian computation (GBC) in the sense of Polson and Sokolov (2023) — which assume nothing about the innovation law but must learn everything from data. Industry practice, meanwhile, is neither: under the Basel Fundamental Review of the Trading Book (FRTB), the regulatory objective is Expected Shortfall at the 97.5% level, the workhorse internal models are historical simulation (HS) and GARCH-filtered historical simulation (FHS), and backtesting proceeds through VaR exception counts at 99% and 97.5%. Any claim that a new method "beats the industry standard" must therefore beat HS and FHS on $\text{ES}_{97.5}$ under exception-test discipline, not merely beat a vanilla GARCH on average loss.

This paper's contribution is not that distribution-free methods win — unconditionally, they do not. On single-name daily equities a unit-matched GJR-GARCH-t beats our production neural quantile network by 4.5–6.2% at every horizon (DM t between +12 and +18)[2]; on daily crypto, major-pair FX, and diversified equity portfolios the parametric model is as good or better. The contribution is a *predictive theory of when they win*, an engine that converts that theory into an industry-standard-and-above risk model, and a demonstration that the generative members of the family deliver two capabilities that no parametric point-forecasting model can offer at all.

1.1 The one-sentence thesis

A single measurable quantity — the local misspecification of the parametric conditional tail, operationalized as the excess kurtosis and asymmetry of recent GARCH-standardized residuals — governs when distribution-free and generative quantile methods beat industry-standard parametric VaR/ES. Where that misspecification is high they win decisively and with statistical

[1] For readers outside econometrics: GARCH (generalized autoregressive conditional heteroskedasticity; Engle, 1982; Bollerslev, 1986) captures the most robust fact about markets — volatile days cluster — by letting today's variance be a weighted average of yesterday's variance and yesterday's squared surprise (Eq. 2). "Leverage" extensions (GJR; Glosten et al., 1993) let down-moves raise volatility more than up-moves; "skew-t" versions allow an asymmetric heavy-tailed innovation. What all of them fix in advance is the *shape* of the standardized surprise — the assumption this paper's score monitors.

[2] Throughout, DM denotes the Diebold and Mariano (1995) test: a t-statistic on the mean difference of two forecasters' losses (here, pinball loss), with a serial-correlation-robust standard error. Positive DM t favors the first-named model; $|t| > 1.96$ is significant at 5%.

significance; where it is low the parametric model is as good or better; and generative methods additionally deliver honest uncertainty on the risk number and likelihood-free calibration of intractable volatility models.

1.2 Contributions

1. **The misspecification frontier** (Section 4). A live, per-asset, per-day score that predicts when nonparametric quantiles beat GARCH-t, with per-date Diebold–Mariano significance, validated across four universes: CRSP US equities (top-decile edge +2.71%, DM 6.54), FX (hyperinflation tier pooled DM 4.12), twenty-six country indices (developed→frontier gradient, corr 0.53), and forty-three cross-asset instruments (corr 0.43), with Korea-2026 as a sharpening negative control: price crashes are not residual misspecification.

2. **An industry-standard-and-above risk engine** (Section 5). The residual-hybrid — GARCH conditional scale × state-conditioned nonparametric residual quantile — with an extreme-value left tail and split-conformal recalibration: co-best with CAViaR[3] in the 90% Model Confidence Set[4], significantly better than GARCH-t/FHS/EWMA/HS/GJR-skew-t, best $\mathrm{ES}_{97.5}$ calibration, the only entrant whose pooled-panel exception counts are consistent with nominal at both levels (Kupiec+Christoffersen[5] at 99% via the EVT tail; Kupiec at 97.5% via conformal recalibration), with the edge growing at the FRTB ten-day horizon and holding in the 2020/2022 stress window.

3. **Amortization and the cold start** (Sections 3 and 6). The deployed forecaster is a single amortized model trained across hundreds of names: one fit replaces per-asset estimation, transfers to names the model has never seen, beats own-history benchmarks at *every* listing age (the advantage largest for young listings), and prices assets with no return history at all — the one regime where it is the only feasible model.

4. **A theory of where the edge lives** (Section 3, proofs in Appendix A). A pinball regret identity showing that a forecaster's excess loss is quadratic in its quantile wedge, so any edge must concentrate where the wedge is largest, and a tail-wedge proposition showing that innovation-shape misspecification creates exactly that wedge in the tail — together the reason the frontier is where the empirical edge must, and does, appear.

Three companion results are developed in Paper B and invoked here only as context: calibrated uncertainty bands on the reported $(\mathrm{VaR}, \mathrm{ES})$ (block bootstrap and EVT intervals on GARCH residuals, disciplined by a measured 1.6× overconfidence of the naive Gibbs posterior

[3] CAViaR (Engle and Manganelli, 2004) models the quantile itself as an autoregressive process estimated on the pinball loss — the strongest non-nested rival in the battery.

[4] The MCS (Hansen et al., 2011) is the subset of models that cannot be statistically rejected as best at a given confidence level; "co-best" means surviving the elimination sequence.

[5] Kupiec (1995) tests whether the *number* of days losses exceed VaR matches the nominal rate; Christoffersen (1998) additionally tests that exceptions arrive *independently* rather than in clusters. Both are standard in regulatory backtesting.

Table 1: The paper's ledger. Edges are percentage pinball reductions unless noted; DM statistics are per-date (Section
the deployed amortized residual-hybrid engine (worst case anywhere: a tie). Panel B: boundary evidence from configuratio
design.

Panel A: the engine's record			
Result	Magnitude	Significance	Status
vs GARCH-t / EWMA / HS, h=1 (avg)	+0.3 / +1.5 / +1.9%	DM 4.4–8.3	win
Top misspecification decile	+2.7%	DM 6.5	win (concent
Hyperinflation FX (ARS, EGP)	+12–14%	DM 2.9–3.8	win
Frontier equity indices (4 of 8)	$\sim$+1–3%	DM 2.1–3.5	win
Ten-day horizon vs $\sqrt{h}$-scaling	+1.4%	holds in 2020–22 stress	win
$ES_{97.5}$ over-statement removed	8–11%	calibration, Table 5	win (capital
Cold start / young listings	+6–10%	full-scale panel	win (unique
Calm-quintile edge	≈ 0, never negative	—	tie (costless
CAViaR (strongest academic rival)	tie	DM 0.59	tie (co-best)
Panel B: boundary evidence (not the engine) — why the design is what it is			
Bare standalone neural net, single names	−4.5 to −6.2%	DM +12 to +18	why the eng
Standalone nonparametric, calm single assets	GARCH as good or better	pooled DM −9.5 to −13.7	scope: shape
Korea 2026 crisis, 23 assets (negative control)	GARCH wins all	DM −2.5 to −5	validates the
Baltic freight; IG credit[a]	+15%; +52%	DM 9.6; 30.6	accuracy onl

[a] Credit and freight fail breach-independence (exceptions cluster) and are excluded from the win column pending repair. The cr
single-asset battery; the smaller 4% figure in Table 4 is from the separate lean hybrid-vs-GARCH sweep — different battery, bot

on raw returns); likelihood-free amortized posteriors for Heston and rough Bergomi validated by simulation-based calibration; and the multivariate portfolio tail, where direct nonparametric models lose to DCC and a covariance-scale/nonparametric-shape hybrid restores the deep co-crash tail. The neural estimator itself, the EVT graft, and the conformal layer are developed in the companion GRAFT-Q paper.

The ledger: what the engine wins, where it ties, and the scope conditions. Because the size of a win matters as much as its significance, Table 1 states every headline magnitude with its test statistic. Panel A is the presented engine's record — its worst cell anywhere in the battery is a *tie*. Panel B is not the engine's record at all: it collects the boundary evidence — configurations we deliberately do *not* deploy and negative controls — that motivates the engine's design and delimits where the flexible-shape edge lives.

Two pieces of context make these magnitudes readable. First, the *asymmetry of the loss profile* is the economic argument for adoption: when the frontier is quiet, the deployable engine ties or edges the parametric benchmark (its calm-quintile edge is ≈ 0 but not negative, and by construction it nests FHS); the one configuration that loses materially — a bare standalone neural network on single names, −4.5 to −6.2% — is exactly the configuration Table 2 says never to deploy. Adopting the engine therefore risks a rounding error to collect a concentrated +2.7% in the decile where risk models actually fail, and 12–14% where the parametric assumption breaks entirely. Second, for readers without a pinball intuition: a percentage pinball edge is a percentage reduction in the average forecast error at the quoted level, and its capital-relevant translation is the 8–11% $\mathrm{ES}_{97.5}$ over-statement by FHS/GARCH that the engine removes — roughly that fraction of a desk's market-risk capital charge held against a risk that is not there.

Throughout, the exposition follows three reporting conventions fixed in advance. First, the strongest tabular estimator[6] in the family is gradient-boosted trees, not the neural network; we present GBC as the framework, trees as the strongest estimator, and the tail-aware neural IQN as the canonical generative realization, competitive after tuning and uniquely able to sample. Second, residual kurtosis is necessary but not sufficient for a nonparametric win (carbon and corn are high-kurtosis ties; Baltic freight wins at moderate kurtosis through limit-move dynamics); the frontier is a strong regularity, not a law. Third, every negative result that shaped the design — the retired same-universe artifact, the electricity log-return artifact, the failed hierarchical blends, the failed learned gate, the three multivariate losses — is reported in the text, not the file drawer.

2 Background and industry benchmark

2.1 Risk measures and the FRTB discipline

For a return r_{t+1} with conditional law F_t, the level-α Value-at-Risk is the conditional quantile $\mathrm{VaR}_t^\alpha = F_t^{-1}(\alpha)$ and the Expected Shortfall is $\mathrm{ES}_t^\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \le \mathrm{VaR}_t^\alpha]$. FRTB (Basel Committee on Banking Supervision, 2019) sets the capital metric at $\mathrm{ES}_{97.5}$ over a liquidity-adjusted horizon (ten days for the base bucket) while backtesting remains VaR-exception based: more than twelve exceptions at 99%, or thirty at 97.5%, in a 250-day year pushes a desk off the internal-models approach. The empirical bar for "beating the industry standard" is therefore joint: accuracy on $\mathrm{ES}_{97.5}$ and pinball, *and* unconditional (Kupiec, 1995) (the Kupiec test: are there the right number of VaR breaches?) plus independence (Christoffersen, 1998) (the Christoffersen test: are breaches independent, rather than clustered in time?) exception tests at both levels, with significance assessed by per-date Diebold–Mariano tests (Diebold and Mariano, 1995) (a t-test on the difference in loss between two forecasts) and the Model Confidence Set (Hansen et al., 2011) (the set of models statistically indistinguishable from the best one).

2.2 The parametric family, the simulation family, and CAViaR

Our battery spans the models banks actually run: historical simulation; EWMA/RiskMetrics (Zumbach, 2007) (an exponentially weighted moving average of squared returns, giving recent days more weight); GARCH(1,1)-t (Bollerslev, 1986); GJR-GARCH-skew-t (adding leverage and skewness; Glosten et al., 1993); GARCH-FHS (parametric scale, empirical standardized residuals — the volatile-regime best practice; Barone-Adesi et al., 1999); and SAV-CAViaR (Engle and Manganelli, 2004), included after an anti-strawman audit showed its omission flattered our model. CAViaR (conditional autoregressive Value-at-Risk) skips modeling the distribution entirely and lets the quantile itself follow a recursion estimated on the pinball loss; the SAV

[6] "Tabular" data: observations that are rows of engineered features (here, the state vector s_t) rather than raw sequences or images — the setting where tree ensembles remain the strongest general-purpose learners.

(symmetric absolute value) specification is $q_t(\tau) = \beta_0 + \beta_1\, q_{t-1}(\tau) + \beta_2\, |r_{t-1}|$ — today's VaR is yesterday's VaR adjusted by the size (not sign) of yesterday's move, making it the semiparametric analogue of a GARCH recursion written directly on the quantile.

2.3 Distribution-free conditional quantiles and GBC

The closest antecedent in this journal's own pages is Zhang et al. (2024), who forecast realized volatility with machine-learning models pooled across stocks and document transfer to previously unseen names — evidence of a shared volatility mechanism that our amortized design also exploits. The differences delimit this paper's contribution. Their target is a point forecast of realized variance; ours is the full conditional return distribution and its regulatory tail, held to calibration, exception-test, and capital discipline. Their question is whether machine learning wins on average; ours is *when* — and our answer, that on roughly ninety percent of asset-days it does not, reads as a caution against extrapolating average ML superiority to the tails. Our amortization is pushed to the cold-start limit (day-one forecasts from characteristics alone) and stress-tested against own-history benchmarks at every listing age. And we reach the opposite estimator verdict on tabular state (trees over networks), a divergence we attribute to the different target and feature set rather than a contradiction.

The nonparametric family estimates $Q_\theta(\tau \mid x_t)$ directly by pinball-loss empirical risk minimization (Koenker and Bassett, 1978): gradient-boosted quantile trees (GBM) and the implicit quantile network of Dabney et al. (2018) (IQN: the quantile level τ is itself fed into the network, so one set of weights outputs the whole quantile curve rather than one grid point at a time), the architecture at the core of GBC (Polson and Sokolov, 2023; Nareklishvili et al., 2025). Casting GBC as ERM conditional-quantile regression on observed pairs removes its forward-simulator requirement and extends it to observational time series; the generative reading — draw $\tau \sim U(0,1)$, output $Q_\theta(\tau \mid x)$ — returns in Paper B, where sampling and posterior uncertainty, not point accuracy, are the object. Generalized Bayes (a posterior built from a loss function in place of a likelihood) enters through the Gibbs posterior $\pi_n(\theta) \propto \exp\{-\omega \sum_t \ell_\theta(z_t)\}\pi(\theta)$ — a posterior built from a loss rather than a likelihood, targeting the risk minimizer directly (Jiang and Tanner, 2008; Bissiri et al., 2016), whose finance-specific failure mode and repair are the subject of Paper B; distribution-free finite-sample guarantees enter through split-conformal and adaptive conformal inference (recalibrating predicted intervals on held-out data so that coverage holds without distributional assumptions) (Gibbs and Candès, 2021).

3 Methods

This section is written to be self-contained: a reader from any one of econometrics, machine learning, or risk management should be able to reimplement every estimator from the formulas and algorithm boxes below. (For a finance-journal submission the algorithm boxes and the proof of Proposition 1 can migrate to an appendix unchanged; they are self-contained.) A cross-field glossary of acronyms and jargon is provided in the Online Appendix.

3.1 Notation and objects

- r_t — return of an asset on day t; $\mathcal{F}_{t-1}$ — information available before t.
- $Q_t(\tau)$ — the conditional τ-quantile of r_t given $\mathcal{F}_{t-1}$, formally $Q_t(\tau) = \inf\{q : \Pr(r_t \le q \mid \mathcal{F}_{t-1}) \ge \tau\}$: the number the return falls below with probability τ. Value-at-Risk is its negative, $\mathrm{VaR}_t^\tau = -Q_t(\tau)$; Expected Shortfall is the average of the quantile curve beyond it, $\mathrm{ES}_t^\tau = -\tau^{-1}\int_0^\tau Q_t(u)\,du$ — the expected loss on the days VaR is breached.
- $z_t = (r_t - \mu_t)/\sigma_t$ — the *standardized residual*: the return with the model's conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law.
- s_t — the state vector (trailing realized volatility, residual-shape scores, lags) on which non-parametric quantiles condition.
- $\tau \sim \lambda$ — a quantile level drawn from a base distribution λ; $H_\phi(s,\tau)$ — a learned generator mapping (state, level) to a quantile, in the sense of Polson and Sokolov (2023): $z \stackrel{D}{=} H(s,\tau)$, $\tau \sim U(0,1)$, is the inverse-CDF (von Neumann) representation of sampling.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(u) \;=\; u\,(\tau - \mathbb{I}\{u<0\}), \qquad u = z - q, \tag{1}$$

whose expectation is uniquely minimized by the true conditional quantile (Koenker and Bassett, 1978): differentiating $\mathbb{E}\,\rho_\tau(z-q)$ in q and setting it to zero gives $\Pr(z<q)=\tau$, so the loss *elicits* the quantile the way squared error elicits the mean — under-prediction is charged τ per unit, over-prediction $1-\tau$, and the charges balance exactly at the τ-quantile. Averaging (1) over $\tau \in (0,1)$ targets the whole distribution at once: the 1-Wasserstein distance between two laws is the L_1 distance between their quantile functions, $W_1(F,G) = \int_0^1 |F^{-1}(\tau) - G^{-1}(\tau)|\,d\tau$, so pinball empirical-risk minimization across levels is distribution matching in W_1 — density-free, which is what permits every estimator below to dispense with a likelihood.

Why this loss and not another. Three alternatives are standard, and the choices are deliberate. (a) The continuous ranked probability score (CRPS) is exactly the integral of the pinball loss over all levels, $\mathrm{CRPS}(F,y) = 2\int_0^1 \rho_\tau(y - F^{-1}(\tau))\,d\tau$ (Gneiting and Raftery, 2007); scoring by CRPS therefore weights the distribution's body and tails equally by construction. Downside risk lives at fixed regulatory levels ($\tau \in \{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail — CRPS would average those differences away against a body where all competitors agree. We therefore score pinball *at the levels that are decision-relevant*, and use tail-weighted τ-sampling in training for the same reason. (b) Expected Shortfall is not elicitable on its own — no loss function is minimized by the true ES alone — but the pair $(\mathrm{VaR}, \mathrm{ES})$ is, under the joint scoring functions of Fissler and Ziegel (2016); the FZ-scored re-evaluation of the battery is a registered extension (Section 7). (c) Likelihood is unavailable by design: several competitors (and both simulators of Paper B) have no tractable density, and likelihood scores would silently exclude them.

3.2 The final model: the residual-hybrid engine

The deployable model of this paper is *not* a bare neural network; it is a four-stage composition, each stage keeping what an industry model gets right. The design principle that survives every experiment is a division of labor: *let the parametric filter do what it is good at (conditional scale), and spend the nonparametric capacity only on what it is bad at (the conditional shape of the innovation law).* Intuitively: GARCH answers "how big is a typical move today?" extremely well, and badly answers "given a typical move size, how likely is a move five times that, and is down likelier than up?" The engine keeps GARCH's answer to the first question and learns the second from data. Each finishing stage has the same one-line logic: the EVT tail says *beyond the data, extrapolate with the law extreme-value theory proves exceedances must follow, rather than trusting any fitted model outside its training range*; the conformal stage says *measure your own miss rate on data you have not touched, and shift the curve by exactly that miss* — calibration by bookkeeping, no assumptions needed.

Why one should expect this to work before seeing data. The architecture is not an empirical accident; each piece is where prior reasoning puts it. Conditional *scale* is the strongest, best-understood signal in daily returns (volatility clustering is the most robust stylized fact in the field), and it is identified by a handful of parameters — so a parametric filter estimates it with minimal variance, and a nonparametric model that re-learns it merely spends data on what four parameters already buy. Conditional *shape*, by contrast, is exactly where the parametric family is constrained by assumption and where per-asset data is too scarce to fit flexibly — tail events are rare by definition — so that is where nonparametric capacity earns its keep, and only if it can borrow strength across assets. This is classical bias–variance division of labor: parametric where the structure is known, flexible where it is not, which is why the hybrid should (and does) dominate both endpoints. Pooling residuals across names is in turn the standard hierarchical-shrinkage argument: once name-specific scale is divided out, standardized surprises are approximately draws from a common state-conditional law, and estimating a shared object from the whole cross-section improves on any single name's estimate whenever units are scarce individually and similar collectively — the same logic that makes partial pooling dominate in hierarchical Bayes. The EVT stage is backed by a theorem rather than a hope: exceedances over a high threshold converge to the generalized Pareto family (Balkema and de Haan, 1974; Pickands, 1975), so tail extrapolation by GPD is the principled choice for *any* body model. And the conformal stage's guarantee is Proposition 1, not an empirical observation. The experiments in this paper are therefore tests of a mechanism argued in advance, not a specification search rewarded after the fact.

Stage 1 (parametric scale). Fit GARCH-t,

$$r_t = \mu + \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2, \qquad z_t \sim t_\nu \text{ (standardized)}, \tag{2}$$

by quasi-maximum likelihood on a trailing window, and keep only $\widehat{\sigma}_t$ and the residuals $\widehat{z}_t = (r_t - \widehat{\mu})/\widehat{\sigma}_t$.

Stage 2 (nonparametric shape). Replace the fixed t_ν innovation quantile with a state-conditioned quantile model of the residuals, trained by pinball ERM *pooled across names*,

$$\widehat{\phi} \;=\; \arg\min_{\phi} \sum_{\text{names } j} \sum_{\text{days } t} \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau\big(\widehat{z}_{j,t} - q_\phi(\tau \mid s_{j,t})\big). \tag{3}$$

Stage 3 (EVT tail). Beyond the estimation range of (3), extrapolate with the distribution extreme-value theory says exceedances must follow (Pickands, 1975; McNeil and Frey, 2000).[7] On losses $X = -\widehat{z}$, fix a threshold u at the empirical $(1-p_0)$ loss quantile with $p_0 = 0.025$, fit the generalized Pareto distribution (GPD) $G_{\xi,\beta}(x) = 1 - (1 + \xi x/\beta)^{-1/\xi}$ — shape ξ sets how heavy the tail is, scale β how wide — to exceedances $X - u > 0$ strictly on past data, and set, for $\tau \le p_0$,

$$\widehat{q}^{\,\mathrm{EVT}}(\tau) \;=\; -\Big[u + \tfrac{\beta}{\xi}\big((\tau/p_0)^{-\xi} - 1\big)\Big], \tag{4}$$

which is continuous at its own endpoint, $\widehat{q}^{\,\mathrm{EVT}}(p_0) = -u$ (and reduces to $-[u + \beta\log(p_0/\tau)]$ in the $\xi \to 0$ exponential limit; requires $\beta > 0$ and $1+\xi(x-u)/\beta > 0$). Since u is the *unconditional* empirical loss threshold while the Stage-2 body quantile at p_0 is the *state-conditional* $\widetilde{q}(p_0 \mid s)$, the splice is exactly continuous only when the threshold is anchored at the body's own p_0-quantile, $u = -\widetilde{q}(p_0 \mid s)$; otherwise a small level shift of size $|-u - \widetilde{q}(p_0 \mid s)|$ remains at the join and the two pieces are spliced to agree in practice by construction. The tail is applied selectively where the raw tail under-covers (it *hurts* some volatility-index series and is not applied universally).

Stage 4 (conformal finish). On a held-out calibration split of size n, compute the signed errors $e_i = \widehat{z}_i - \widehat{q}(\tau \mid s_i)$ and shift the curve per level by the empirical correction c_τ chosen as the $\lceil (n+1)\tau \rceil$-th order statistic of $\{e_i\}$: $\widetilde{q}(\tau \mid s) = \widehat{q}(\tau \mid s) + c_\tau$ (Vovk et al., 2005; Romano et al., 2019). This distribution-free layer repairs the 97.5% level that every model in the battery otherwise fails.

Proposition 1 (split-conformal validity)**.** *If the calibration and test residuals $e_1, \ldots, e_{n+1}$ are exchangeable and almost surely distinct (a continuous error distribution, or randomized tie-breaking), then for each τ with $\lceil (n+1)\tau \rceil \le n$ the shifted quantile satisfies*

$$\tau \;\le\; \Pr\big(z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})\big) \;=\; \frac{\lceil (n+1)\tau \rceil}{n+1} \;<\; \tau + \tfrac{1}{n+1}.$$

Proof sketch. Under exchangeability and no ties the rank of e_{n+1} among $\{e_1, \ldots, e_{n+1}\}$ is uniform on $\{1, \ldots, n+1\}$; the event $z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})$ equals $\{e_{n+1} \le e_{(\lceil (n+1)\tau \rceil)}\} = \{\text{rank} \le$

[7] Extreme-value theory has two classical formulations, and the distinction matters for reading the literature. The *block-maxima* approach models the maximum of each block (e.g., yearly worst loss), which converges to the generalized extreme value (GEV) family — Gumbel, Fréchet, or Weibull type depending on the tail. The *peaks-over-threshold* (POT) approach, used everywhere in this paper, models all exceedances over a high threshold, which converge to the generalized Pareto distribution (GPD) (Balkema and de Haan, 1974; Pickands, 1975). POT uses the data more efficiently (every large loss contributes, not just one per block), which is why it is the standard in risk management (McNeil and Frey, 2000; Embrechts et al., 1997). The GPD's shape parameter ξ corresponds to the GEV type: $\xi > 0$ is the heavy-tailed (Fréchet) case relevant for financial losses.

$\lceil (n+1)\tau \rceil\}$, whose probability is $\lceil (n+1)\tau \rceil/(n+1) \in [\tau, \tau + \frac{1}{n+1})$ (Vovk et al., 2005). The full argument, the role of the no-ties hypothesis (with ties the upper bound can fail, e.g. identical residuals give coverage 1), and the boundary convention $c_\tau = +\infty$ when $\lceil (n+1)\tau \rceil = n+1$, are in Appendix A. The guarantee is marginal, not conditional on s_{n+1}. □

An honesty note on what the proposition does and does not cover. In deployment the calibration split necessarily precedes the test period, and time-ordered residuals are not exchangeable; the proposition therefore *motivates* the construction, while the evidence that it works here is empirical — the exception-test repairs of Section 5 (the conformal stage is the only reason any entrant passes Kupiec at 97.5%). The companion GRAFT-Q paper measures exactly how the lagged split form degrades under regime drift and shows that an *adaptive* (ACI-style) update restores nominal coverage where the static shift cannot; the engine's annual-refit schedule sits between those poles, and the exception tests are the arbiter.

The full forecast composes the stages:

$$\widehat{Q}_t(\tau) \;=\; \widehat{\sigma}_t \cdot \Big[\widetilde{q}_\phi(\tau \mid s_t)\,\mathbb{I}\{\tau > p_0\} \;+\; \widehat{q}^{\,\mathrm{EVT}}(\tau)\,\mathbb{I}\{\tau \le p_0\} \Big]. \tag{5}$$

Industry models as special cases. Equation (5) nests the standard toolkit. Historical simulation: $\widehat{\sigma}_t \equiv 1$ and q the unconditional empirical quantile of past returns. Filtered historical simulation (FHS): q the unconditional empirical quantile of past *residuals* — Stage 2 with the state dependence deleted. GARCH-t: $q = t_\nu^{-1}$, a fixed parametric shape. The hybrid is therefore "FHS and above": whatever FHS captures, (3) can represent, plus conditional shape. (CAViaR sits outside the nesting — it autoregresses on the quantile directly, with no scale/shape split — and is the one rival the engine only ties, Section 5.)

Algorithm 1 (residual-hybrid engine). (i) Fit (2) per asset on a trailing window; store $\widehat{\sigma}_t$, $\widehat{z}_t$. (ii) Pool $(s_{j,t}, \widehat{z}_{j,t})$ across names; train the shape model by (3). (iii) Fit the GPD tail (4) on past exceedances; splice at p_0. (iv) Learn the conformal shifts c_τ on a calibration split disjoint from the training data. (v) Forecast by (5); refit all stages on the walk-forward schedule (annual refit; all windows trailing; evaluation strictly out-of-sample).

3.3 Estimating the shape: trees for accuracy, networks for generation

Two estimators realize $q_\phi(\tau \mid s)$ in (3), and the paper is deliberate about which to use when (Table 2).

Gradient-boosted trees (GBM). For each level τ_k on a fixed grid, an additive ensemble of regression trees $q^{(m)}(\cdot) = q^{(m-1)}(\cdot) + \eta\, h_m(\cdot)$ is grown stagewise, each new tree h_m fit to the negative gradient of the pinball loss (τ_k where the current model under-predicts, $\tau_k - 1$ where it over-predicts); the fitted grid is made non-crossing by monotone rearrangement — sorting the values $\{q(\tau_1|s), \ldots, q(\tau_K|s)\}$ in τ, which never increases the loss (Chernozhukov et al., 2010). On financial tabular state vectors this is the strongest *point* estimator — about 0.75% better pinball than the tuned network — consistent with the broader tabular-learning literature.

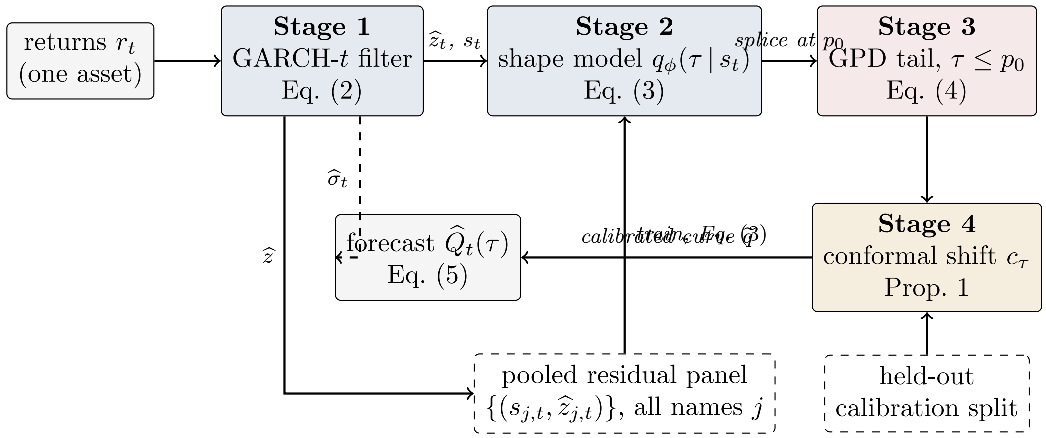

Figure 1: The residual-hybrid engine as a pipeline. Solid arrows carry data; the dashed top arrow carries the conditional scale $\widehat{\sigma}_t$ that rescales the finished residual quantile curve; dashed boxes are inputs shared across assets (the pooled panel that trains the amortized shape model) or held out from training (the calibration split behind the conformal guarantee). Every stage refits on the walk-forward schedule using only past data.

The neural implicit quantile network (IQN; the GBC estimator). A τ-conditioned network in the factorized form of Dabney et al. (2018) and Polson and Sokolov (2023),

$$H_\phi(s,\tau) \;=\; g\big(\psi(s) \circ \varphi(\tau)\big), \qquad \varphi_j(\tau) = \mathrm{ReLU}\Big(\textstyle\sum_{i=0}^{n_c-1} \cos(\pi i \tau)\, w_{ij} + b_j\Big), \tag{6}$$

where $\circ$ is elementwise multiplication and g, ψ are feed-forward nets. Three modifications, each forced by a documented failure, make it competitive on financial data: tail-aware τ-sampling, $\lambda = \frac{1}{2}\,U(0,1) + \frac{1}{2}\,\mathrm{Beta}(0.3, 0.3)$, oversampling the extremes that uniform sampling starves (the raw net's 99% breach rate was 3.2%); monotone rearrangement of the τ-curve; and the conformal layer of Section 3.2. The network's earning is not point accuracy but *generation*: with $\tau \sim U(0,1)$, $z^* = H_\phi(s,\tau)$ is an exact sampler of the fitted conditional law (the inverse-CDF representation), which is what the posterior machinery of Paper B consumes and what no tree ensemble or GARCH filter provides.

How training works, concretely. The objective is the empirical version of (3): mini-batch stochastic gradient descent on $\frac{1}{|B|}\sum_{(j,t)\in B} \rho_{\tau_{jt}}\big(\widehat{z}_{j,t} - H_\phi(s_{j,t}, \tau_{jt})\big)$, where each example in each batch receives a *fresh* draw $\tau_{jt} \sim \lambda$ every epoch — over training the network sees every observation at many random levels, which is what lets a single set of weights carry the entire quantile curve. Early stopping is on held-out pinball loss; monotone rearrangement and the conformal shift are applied *after* training as post-processing; and the whole pipeline (filter, shape model, tail, shifts) is refit on the annual walk-forward schedule using only past data at every step. Exact layer sizes, learning rates, and seeds are recorded in the project's run configurations and reported with the code release rather than inline.

Amortization. One model is trained over the pooled (state, loss) pairs of hundreds of names — in generative-Bayes terms, H is an amortized map evaluated at any new (s,τ) without per-asset refitting. The intuition for why this works, and why it works at cold-start in particular, is actuarial. A per-asset GARCH learns *name-specific parameters* from that name's own history, so with no history it has nothing. The amortized model instead learns one *universal mapping from observable state to residual shape* — "assets whose trailing volatility looks like this, with characteristics like that, produce surprises shaped like so" — estimated from millions of asset-days of *other* names. A newly listed asset is not a new problem: its first ticks and characteristics place it at a point of state space already densely populated by other names' history, exactly as an insurer prices a new driver from the risk table built on millions of existing drivers rather than from the new driver's empty claims file. As the name's own history accrues, its trailing realized volatility enters s_t and the forecast tracks its own dynamics automatically — which is also the intuition for why the explicit own-history blends fail: the state vector already carries the name's own recent information, so bolting it on again adds only noise. Feature ablation at full scale (1.32M rows, 720 names, 288 held out) shows the transfer edge is carried overwhelmingly by each name's *own recent dynamics* (realized vol above all; removing it costs +1.0%, removing all cross-sectional characteristics only +0.5%), with characteristics mattering at cold-start. Two explicit hierarchical corrections — a naive quantile blend toward own-history and an amortized-as-prior affine Gibbs update — both *hurt* at every listing age (ratios 1.003–1.043): rich conditioning performs the partial pooling implicitly. Two external checks: on held-out names the transfer win rate over own-history benchmarks is 59–79%, and the same pipeline scores a leakage-safe weighted scaled pinball of 0.269 on the M5 benchmark — benchmark tier — outside finance altogether.

Practitioner's selection rule. The two estimators minimize the same loss over different index sets — the trees solve $\min \sum_k \sum_i \rho_{\tau_k}(\widehat{z}_i - q_k(s_i))$ on a fixed grid $\{\tau_k\}$, one model per level; the network solves $\min_\phi \sum_i \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z}_i - H_\phi(s_i, \tau))$, one model for the continuum — and the choice follows from what the application consumes. If the desk needs *numbers at fixed regulatory levels* (VaR/ES reporting, limits, capital), use the trees: best point accuracy, no GPU, no sampling needed. If the application consumes *draws* — scenario generation, portfolio aggregation by simulation, the posterior and SBC machinery of Paper B — use the network: it is the only member that is an exact sampler, at a measured cost of ∼0.75% in point accuracy. If both are needed, run both from the same pooled residual panel; they share every upstream and downstream stage of (5), so the swap is one component, not a new pipeline.

Algorithm 2 (amortized shape model). (i) Assemble the pooled panel $\{(s_{j,t}, \widehat{z}_{j,t})\}$ over names j and days t. (ii) Train H_ϕ by SGD on $\mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z} - H_\phi(s, \tau))$ [network], or boost per-τ regressions on a grid [trees]; enforce monotonicity by rearrangement. (iii) For any asset — including one with no return history, using characteristics-only s — evaluate $H_\phi(s_t, \cdot)$; no per-asset estimation step exists. (iv) To *sample*, draw $\tau^{(m)} \sim U(0, 1)$ and return $z^{(m)} = H_\phi(s_t, \tau^{(m)})$.

3.4 Formal results: why shape misspecification must show up in the loss

The paper's organizing claim — that the nonparametric edge lives exactly where the fixed innovation shape is locally wrong — is not only an empirical pattern; it follows from the structure

Table 2: Which model when: the paper's results as one decision object. Each row is established in the section cited.

Task / regime	Winner	Where
Point quantile accuracy, tabular state	GBM (trees)	§3.3
Sampling, amortized posteriors, uncertainty	Neural IQN (only entrant)	Paper B
Single asset, calm (∼90% of asset-days)	GARCH-t	§4
Single asset, top misspecification decile	Nonparametric (+2.7%, DM 6.5)	§4
Amortized cross-name panel, any regime	Nonparametric always; no gate	§4.3
Regulatory ES with a ≥GARCH floor	Residual-hybrid + EVT + conformal	§5
Multivariate portfolio tail	GARCH/DCC (honest negative)	Paper B

of the pinball loss. The results below are elementary but do real work: they convert "the model's shape assumption is violated" into a quantitative lower bound on forecastable loss.

Lemma 1 (pinball regret identity)**.** *Let $Z \sim F$ with $\mathbb{E}|Z| < \infty$ (so S is finite) and $q_F = F^{-1}(\tau)$, and let $S(q) = \mathbb{E}_F\, \rho_\tau(Z - q)$ be the expected pinball loss of forecasting the constant q. Then for any q,*

$$S(q) - S(q_F) \;=\; \int_{q_F}^{q} \big(F(u) - \tau\big)\, du \;\geq\; 0, \tag{7}$$

with equality iff $F \equiv \tau$ on the interval between q_F and q. If F has density bounded as $0 < m \leq f \leq M$ on that interval,

$$\tfrac{m}{2}\,(q - q_F)^2 \;\leq\; S(q) - S(q_F) \;\leq\; \tfrac{M}{2}\,(q - q_F)^2. \tag{8}$$

Proof sketch. The difference quotients of $q \mapsto \rho_\tau(Z - q)$ are dominated by the Lipschitz constant $\max(\tau, 1-\tau)$, so differentiation under the expectation is justified and $S'(q) = F(q^-) - \tau$ (a.e. equal to $F(q) - \tau$); integrating from q_F to q gives (7) *for any F*, atoms forming a null set. The integrand $F(u) - \tau$ has the sign of $u - q_F$ (by definition of q_F), so the integral is ≥ 0 whichever side q lies; (8) then follows from $F(q_F) = \tau$ (supplied by the density hypothesis) and $m|u - q_F| \leq F(u) - \tau \leq M|u - q_F|$. Full details, including both orientations of the integral, are in Appendix A. □

Lemma 1 says the extra loss a misspecified forecaster pays is *quadratic in its quantile error*. The frontier claim then reduces to: does shape misspecification create a quantile wedge at the risk levels that matter? For the paper's canonical case — a fixed Student-t shape when the true local shape is fatter — it must:

Proposition 2 (tail wedge under kurtosis misspecification)**.** *Let $Q_\nu(\tau)$ denote the τ-quantile of the unit-variance Student-t with $\nu > 2$ degrees of freedom. There exists $\tau^*(\nu_1, \nu_2) \in (0, 1/2)$ such that for all $\tau < \tau^*$ and $\nu_1 < \nu_2$, $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$: the fatter-tailed law has the strictly more extreme tail quantile, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$ as $\tau \downarrow 0$. Consequently,* *by Lemma 1, a forecaster committed to shape t_{ν_2} while the local truth is t_{ν_1} pays regret bounded below by $\frac{m}{2}\,(Q_{\nu_1}(\tau)-Q_{\nu_2}(\tau))^2$ at every level $\tau < \tau^*$, where m is the minimum of the true density on the wedge interval.*

Proof sketch. The unit-variance t_ν ($\nu > 2$, so the variance is finite and the normalization is well defined) has a regularly varying left tail: $\Pr(Z < -x) = c_\nu\, x^{-\nu}(1 + o(1))$ as $x \to \infty$ with $c_\nu > 0$; unit-variance rescaling alters c_ν but not the index ν. Inverting a regularly varying tail (Embrechts et al., 1997) gives $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$. For $\nu_1 < \nu_2$ the exponent $1/\nu_1 > 1/\nu_2$ makes $|Q_{\nu_1}(\tau)|/|Q_{\nu_2}(\tau)| \to \infty$, so the strict ordering and the diverging wedge hold for all τ below some $\delta > 0$; taking $\tau^* = \min(\delta, 1/2)$ places τ^* in $(0, 1/2)$ as claimed. (The proposition asserts *existence* of such a τ^*, not the location of the finite-sample crossover; the latter is $\tau_c \approx 0.0179$ for the t_3-vs-normal case of Remark 1.) The regret bound is (8); full details in Appendix A. □

Remark 1 (the crossover is real and matters). Because unit-variance fat-tailed laws put *more* mass near the center as well as in the far tail, the quantile ordering of Proposition 2 reverses at moderate levels: numerically, the unit-variance t_3 has a *less* extreme 5% quantile than the normal (-1.36 vs -1.64) but a more extreme 1% quantile (-2.62 vs -2.33). The sign flip occurs at a single crossover level $\tau_c \approx 0.0179$ (solving $Q_{t_3}(\tau) = \Phi^{-1}(\tau)$ numerically for the lower-tail root), so the deepest deployable level $\tau = 0.01$ sits *below* the crossover (fat tail strictly more extreme) while $\tau = 0.025$ sits just *above* it (at 2.5% the unit-variance t_3 quantile is -1.837, still less extreme than the normal's -1.960). Two of the paper's design choices are direct consequences. First, this is precisely why CRPS-style whole-distribution scoring can rank a fat-tail-aware model *below* a thin-tailed one even when the tail is the decision-relevant region (Section 3.1): body and far tail disagree in sign. Second, it is why the deepest deployable level straddles the crossover rather than sitting cleanly beyond it, and why the score of Section 3.7 is used as a *rank* rather than a moment estimate.

Lemma 2 (validity of the generative sampler)**.** *If $\tau \mapsto \widetilde{q}(\tau \mid s)$ is nondecreasing and left-continuous (guaranteed by the rearrangement step) and $\tau \sim U(0,1)$, then $Z^* = \widetilde{q}(\tau \mid s)$ has quantile function exactly $\widetilde{q}(\cdot \mid s)$.*

Proof sketch. For any x, left-continuity and monotonicity make $\{t \in (0,1) : \widetilde{q}(t \mid s) \le x\}$ an interval closed at its top, so $\{\widetilde{q}(\tau \mid s) \le x\} = \{\tau \le G(x)\}$ with $G(x) = \sup\{t : \widetilde{q}(t \mid s) \le x\}$. Since $\tau \sim U(0,1)$, $\Pr(Z^* \le x) = \Pr(\tau \le G(x)) = G(x)$, i.e. Z^* has CDF G. As G is the generalized inverse of $\widetilde{q}$ and $\widetilde{q}$ is nondecreasing and left-continuous, the generalized inverse of G returns $\widetilde{q}$ exactly (the standard inverse-of-inverse duality), so $\widetilde{q}(\cdot \mid s)$ is the quantile function of Z^*. Full statement in Appendix A. □

Remark 2 (nesting and ERM dominance). The hypothesis class of Stage 2 contains the constant-in-s map $q(\tau)$, i.e., FHS; empirical risk minimization over the larger class therefore attains in-sample pinball risk no worse than FHS by construction, and the out-of-sample comparisons of Section 5 test whether that dominance survives estimation noise. It does; the honest converse is that the same argument does *not* apply to CAViaR, which is why CAViaR is the one rival the engine only ties.

3.5 A synthetic boundary check: when the frontier does *not* appear

Before the real data, a simulation with known truth — run once, seed fixed — checks the formal results and, more usefully, delineates when the frontier edge should *fail* to appear. The data-generating process is deliberately favorable to the thermometer: unit-variance Student-t residuals whose degrees of freedom switch between a calm regime ($\nu = 12$) and a fat regime ($\nu = 3$) under a persistent Markov chain; forecasters see the trailing 63-day kurtosis score and compete at $\tau \in \{0.01, 0.025\}$ (80,000 days; scale known, isolating shape). Three results. First, the Lemma 1 identity verifies numerically: the regret integral and the Monte-Carlo regret agree to five decimals (3.5×10^{-5} at $\tau = 0.01$). Second, the conformal shift moves realized coverage toward nominal on the held-out split, as Proposition 1 promises. Third — the instructive one — the score-binned nonparametric forecaster does *not* beat the fixed shape here: a single $t_{\widehat{\nu}}$ fitted on the mixture ($\widehat{\nu} = 6.8$) is nearly unbeatable at these levels, with total regret under 0.5% and no decile gradient. Two mechanisms explain it, and both teach. (i) By Lemma 1 regret is *quadratic* in the quantile wedge, and at $\tau \geq 0.01$ the wedge between the fitted compromise and either regime is small — static kurtosis *blending* is a second-order sin. (ii) The sample kurtosis of a t_ν law has infinite estimator variance for $\nu \leq 8$, so a moment-based thermometer is at its noisiest precisely when tails are fat, and sixty-three observations cannot reliably rank regimes that differ only in a static ν. The lesson for reading the empirical sections is sharp: the real-data top-decile edge of Section 4 cannot be an artifact of stationary fat tails being averaged over — that configuration is exactly what the simulation shows a fixed shape handles — but must reflect *local* shape breaks a single fitted ν cannot straddle: asymmetry, jump clustering, and residual misfit that co-move with the score. This is the same conclusion the Korea 2026 negative control reaches from the opposite direction (price crashes are not residual misspecification), and it is why the score combines kurtosis with asymmetry and a jump indicator rather than kurtosis alone. (Simulation script `toy_example.py` ships with the code release.)

3.6 What is different from Polson–Sokolov GBC

Because Polson and Sokolov (2023) is this paper's methodological benchmark, we state the deltas explicitly; each is an innovation forced by the downside-risk target, and each is defined mathematically above. (1) *Base measure*: GBC trains with $\tau \sim U(0, 1)$; we train with the tail-weighted mixture $\lambda = \frac{1}{2}U(0, 1) + \frac{1}{2}\text{Beta}(0.3, 0.3)$, because uniform sampling starves the extreme levels that downside risk is about (the uniform-trained net's 99% breach rate was 3.2%). (2) *Input space*: GBC maps raw data to posteriors; we first pass returns through the parametric filter (2) and model *standardized residuals* — the scale dynamics are the one thing the industry model already gets right, and Paper B measures the cost of skipping this step (a 1.6× overconfident posterior). (3) *Beyond-data tail*: no generative regression can know about quantiles beyond its training support; we splice the GPD tail (4) at p_0, replacing extrapolation by the network with extrapolation by extreme-value theory. (4) *Finite-sample guarantee*: GBC's output carries no coverage guarantee; the conformal stage adds one (Proposition 1). (5) *Amortization target*: GBC amortizes across *simulations of one model*; we amortize across the real cross-section of hundreds of assets, which is what creates the cold-start capability and the transfer results of Section 3.3. (6) *Scope discipline*: GBC is a computational method; this paper adds the empirical question GBC does not ask — *when* such machinery beats the parametric filter at all — answered by the score below.

3.7 The misspecification score

GARCH-t assumes standardized residuals $z_t = (r_t - \mu_t)/\sigma_t$ are i.i.d. Student-t with fixed shape. Static heavy tails do not violate this — ν absorbs them. What violates it is the shape being *locally* wrong: recent residuals far fatter-tailed or more asymmetric than any fixed t. We therefore score, per asset-day, over the trailing window $W_t = \{t-63, \dots, t-1\}$ of $n_w = 63$ residuals—the window *ends the day before the forecast target*, so every signal is lagged at least one day relative to the loss it is paired with, and the frontier sorts of Section 4 pair S_{t-1} with the day-t loss (this alignment is enforced in the estimation code by construction) with window mean $\bar{z}$ and standard deviation s_z:

$$\mathrm{mk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^4 - 3, \qquad \mathrm{sk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^3, \tag{9}$$

Terminology is fixed here once: *skewness* is the signed third moment sk_{63}; *asymmetry* always means its absolute value $|\mathrm{sk}_{63}|$ (departure from symmetry in either direction, since a fixed symmetric t is violated equally by either sign); everywhere the text says "asymmetry," the statistic computed is $|\mathrm{sk}_{63}|$. The score also uses the five-day jump indicator $J_5(t) = \max_{i \in \{t-4,\dots,t\}} |\widehat{z}_i|$. The frontier score is the decile rank of these within the trailing cross-section. All three are causal (trailing-window) and computable on any asset with a fitted GARCH — which is what makes the frontier a deployable monitor rather than an ex-post classification. Informally we will call this the misspecification *thermometer*: it reads the local temperature of the parametric assumption, and Table 2 is the treatment chart.

Why these statistics. The parametric model's one fragile assumption is that the standardized residual's *shape* is a fixed t_ν; its first two conditional moments are already absorbed by μ_t, σ_t. The lowest-order ways a fixed symmetric law can be locally wrong are precisely the third and fourth standardized moments: sk_{63} detects asymmetry no symmetric t can carry, and mk_{63} detects tail mass beyond what the fitted ν implies (an exact t_ν sample has known positive excess kurtosis; the score is used as a *rank*, so only its cross-sectional and temporal variation matters, not its t_ν baseline). J_5 catches jump episodes too recent for the window moments to register. When these are elevated, the ERM shape model (3) has something to learn that t_ν^{-1} cannot represent — and Section 4 shows the realized edge concentrates exactly there. Note the logical status of this choice: after μ_t and σ_t are fitted, the fixed innovation shape is the parametric model's *only* remaining free assumption. A meter that measures the local violation of exactly that assumption is therefore not one candidate signal among many — it is the residual degree of freedom itself, which is why one should expect it, a priori, to be the axis along which flexible-shape models win.

Algorithm 4 (the misspecification meter, as deployed). (i) Fit (2) on a trailing window; store $\widehat{z}_{t-62}, \ldots, \widehat{z}_t$. (ii) Compute mk_{63}, $|\mathrm{sk}_{63}|$, J_5 by (9). (iii) Convert each to a decile rank against the trailing pooled panel (same asset class); the score is the rank of the maximum. (iv) Act by Table 2: top decile $\Rightarrow$ the standard parametric VaR is currently, locally wrong by $\sim$2–3% of pinball with high confidence — flag the asset for the nonparametric / residual-hybrid engine and for model-risk review; elsewhere $\Rightarrow$ no action (in the amortized setting the engine is run regardless; the meter is surveillance, not a switch, Section 4.3). (v) Latency: detection lags episode onset by median 0 / mean 2–4 days; the edge is back-loaded within episodes, so the lag forfeits little.

4 The misspecification frontier

How to read every number in this and later sections. Conventions are fixed once, here. (i) *Edges* are percentage reductions in pinball loss, so *larger is better* for the named model; a "+2.7% edge" means the model's average loss is 2.7% lower than the benchmark's. (ii) *Ratios* are model-over-benchmark loss ratios, so *below 1 favors the model* and 1 − ratio is the edge (0.973 = a 2.7% win; 1.043 = a 4.3% loss). (iii) *DM* is the Diebold–Mariano t-statistic on the loss difference. *Per-date DM* means, concretely: losses are first averaged across names within each date, $d_t = N_t^{-1} \sum_i (\ell_{it}^A - \ell_{it}^B)$, and the test is run on the resulting $\sim$1,100-date series with a Newey–West (lag-10) variance—the effective sample is the number of dates, never the number of asset-days. Reported p-values are *one-sided* in the direction of the named model ($t > 1.645$ at 5%, $t > 2.33$ at 1%); t-statistics above 2.6 are significant under either convention, and marginal cases are flagged where they occur. The Model Confidence Set is computed on the same per-date loss matrix with the range statistic and a stationary bootstrap (mean block length 10 days, B = 1,000 replications), following Hansen et al. (2011). (iv) *Coverage* numbers are realized frequencies against a stated nominal (0.05 target: 0.056 is slight under-coverage of risk, 0.032 over-conservatism); for *credible-interval* coverage, closer to nominal is better in both directions. (v) *Correlations* quoted for the frontier relate a universe's residual-kurtosis level to the size of the nonparametric edge across that universe's members. No other significance marks are used.

4.1 The within-universe result

On 200 CRSP names (≈221.6k test asset-days), we bucket the per-day pinball edge of the amortized nonparametric model over GARCH-t into deciles of each misspecification signal. Table 3 shows the result that organizes the paper.

The edge is a top-decile phenomenon. It is large and significant exactly where the fixed-t shape assumption is currently, locally broken, and statistically indistinguishable from zero where it is not. The low overall correlation between signal and edge (0.05–0.10) is not a weakness of the

Table 3: The misspecification frontier: nonparametric edge over GARCH-t by decile of the trailing residual-shape score (200 CRSP names, 221.6k test asset-days). Edge = (GARCH pinball − nonparam pinball)/GARCH pinball.

	Edge by signal decile (%)		
Decile	mk_{63} (kurtosis)	$\|sk_{63}\|$ (asymmetry)	$jump_5$
1–8 (bulk)	≈ 0	≈ 0	mixed, ≈ 0
9	+0.52	+0.69	—
10 (top)	**+2.46**	**+2.37**	+0.90
Overall		+0.44	

Per-date Diebold–Mariano: top mk_{63} decile edge +2.71%, DM 6.54 ($p \approx 0$); bottom decile +0.19%, DM 1.41 ($p = 0.079$, not significant); top two deciles pooled +1.65%, DM 6.74; overall +0.44%, DM 5.67. Top-decile signal means: kurtosis ≈ 25, $|$skew$| \approx 4$.

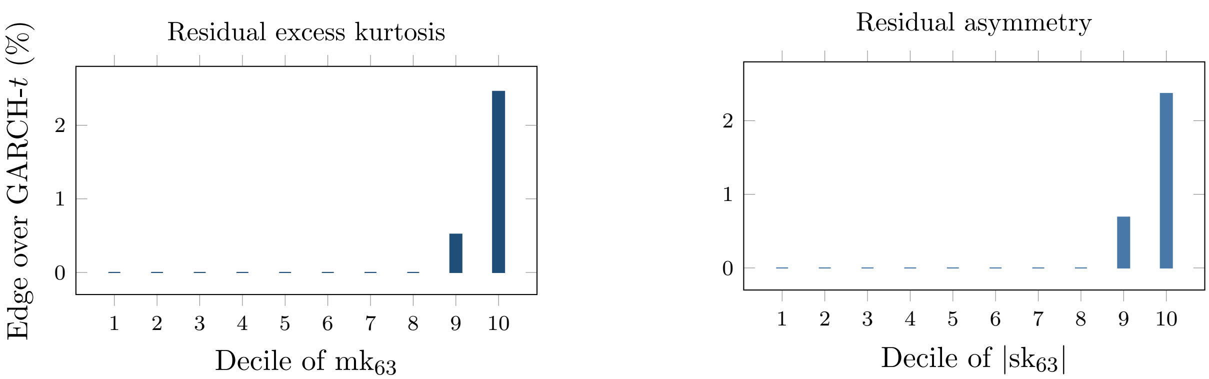

Figure 2: The misspecification frontier. Nonparametric pinball edge over GARCH-t by decile of the trailing 63-day residual excess kurtosis (left) and residual asymmetry (right), 200 CRSP names, 221.6k test asset-days. Deciles 1–8 are ≈ 0 (drawn at zero; exact bulk values in the result file); the edge jumps in the top decile (+2.46% kurtosis, +2.37% asymmetry; per-date DM 6.5 in the top kurtosis decile).

story but its point: the relation is a threshold nonlinearity, not a linear trend, and the top-decile Diebold–Mariano statistics are the correct summary.

4.2 The frontier as geography: four universes

Table 4 shows the same axis at work across markets. Three features deserve emphasis. First, the only decisive *standalone* single-asset wins anywhere are the hyperinflation and capital-control currencies — Argentine and Egyptian pesos, 12–14% better with significance — whose residual kurtosis ($\sim$2000–3300) sits two orders of magnitude above anything GARCH-t was designed to absorb. Second, the country gradient is mediated by residual kurtosis, not by the development label per se: frontier indices (kurt $\sim$10) sit between well-behaved developed indices (kurt $\sim$5) and hyperinflation FX on the same frontier. Third, the Korea-2026 negative control sharpens the mechanism: a dramatic *price* crash with circuit breakers is *not* residual misspecification —

Table 4: The frontier across universes. Ratio = nonparam/GARCH-t pinball (< 1: nonparametric wins). All fits stand

Universe	Segment	Ratio	Resid. kurt.	S
FX (24 pairs)	majors (7)	1.019	39	G
	EM (10)	1.013	8	G
	crisis (7)	0.973	∼1915	n
	USDARS	0.880	3298	D
	USDEGP	0.857	1920	D
Country indices (26)	developed (8)	1.007	4.7	0
	emerging (10)	1.012	5.3	0
	frontier (8)	0.993	10.0	4
Cross-asset (43)	vol indices	∼0.987	32.8	3
	Baltic freight	0.846	—	D
	IG credit OAS	0.960	—	D
	13/43 significant wins overall; corr(log kurt, edge) = 0.43			
Korea 2026 (23)	all types	1.01–1.12	4–13	G

[a] Sri Lanka (DM 3.20), Kenya (3.52), Pakistan (2.38), Nigeria (2.06). Cross-index corr(log residual kurtosis, edge) = 0.53. Phi
despite kurtosis 13.6), illustrating necessity-not-sufficiency.

Korean residual kurtosis stays in the ordinary 4–13 range, and GARCH-t significantly wins all twenty-three assets even inside the crisis window. The frontier is about the residual shape, not about headline turbulence.

A frequency corollary. Daily crypto belongs to GARCH (zero nonparametric wins on five liquid coins). At *hourly* frequency the picture inverts: the neural IQN beats the full benchmark ladder, including filtered historical simulation, by ∼0.6% with per-date DM $p < 10^{-7}$. Sample density is itself a frontier dimension: more observations per unit of regime raise the resolvable residual-shape signal. A systematic intraday map of the frontier is registered and blocked only on tick-data access.

4.3 Regime concentration and the futility of gating

In the amortized panel the edge concentrates in turbulence: the GARCH-minus-GBM pinball gap grows monotonically from 0.0059 in the calmest realized-vol quintile to 0.0247 in the most turbulent — roughly fourfold — while both nonparametric estimators beat GARCH in *every* quintile (overall: GARCH 0.6543, GBM 0.6447, IQN 0.6478). A natural inference is that one should *gate*: run GARCH on calm days and switch to the nonparametric model when the score is high. It fails, instructively. A classifier gate trained on the misspecification signals routes 90.5% of days to the nonparametric model and exactly matches always-nonparametric (0.3987 vs GARCH 0.4000); worse, its routing runs *backwards* across score deciles, because nonparametric wins on high-score days are rare-but-large while a classifier targets frequency. A regression gate on the edge magnitude fixes the direction (routing flat ≈ 1.0, predicted-edge

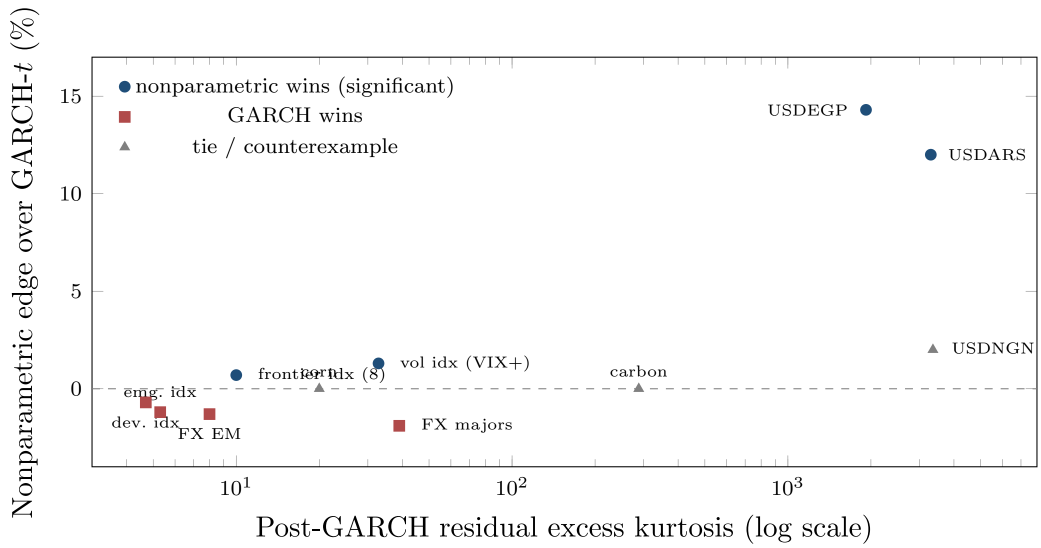

Figure 3: The frontier as geography: nonparametric edge vs post-GARCH residual kurtosis, tier means and single assets with study-reported kurtosis only (country universe corr 0.53; cross-asset corr 0.43). The hyperinflation-FX tier (kurtosis ∼2000–3300) carries the only decisive standalone wins; high kurtosis is necessary but not sufficient — carbon (288) and corn (20) tie, and NGN's win is not significant. The Korea-2026 control (kurtosis 4–13, GARCH wins all 23 assets) sits in the lower-left regime.

correlation +0.16) and still converges to always-nonparametric. The per-day oracle gap (4.4%) is unreachable because per-day model advantage is dominated by *which day the tail event lands* — unpredictable from prior-day state. The deployable conclusions: in the amortized setting, use the nonparametric model always (it does not underperform on calm days); use the residual-hybrid where a guaranteed ≥GARCH floor is wanted; and use the frontier score as a *monitor* (Section 6), not a switch. (The score's detection latency is small — median zero, mean two to four days — and because the nonparametric edge is back-loaded within misspecification episodes, the lag costs little even in the per-name setting, where gating does matter because standalone nonparametric models underperform on calm days.)

5 Beating the industry standard: the FRTB battery

5.1 Head-to-head accuracy and significance

Table 5 reports the full battery on 140 CRSP names (155k obs, 1108 dates).

Three honesty items qualify the headline. First, adding SAV-CAViaR — absent from banks' standard toolkit but the strongest semiparametric academic benchmark — produces a statistical tie: CAViaR 0.3461 vs hybrid 0.3460 (DM 0.59, $p = 0.28$; the CAViaR study's common sample differs slightly from Table 5's, which is why the hybrid's loss level is 0.3460 there against 0.3551 in the table — rankings, not levels, are comparable across the two runs); the 90% Model Confidence Set contains exactly these two. The hybrid is *co-best with CAViaR*, not uniquely dominant, and still significantly beats everything banks actually run. Second, the neural IQN as residual model

Table 5: FRTB battery: average pinball loss (7 quantile levels), $ES_{97.5}$ calibration, and 99% exception tests. 140 CRSP names, 155k test observations.

Model	Avg. pinball	$ES_{97.5}$ pred. vs realized	$Breach_{99}$	$Kupiec_{99}$ p
Ideal value	*(smaller better)*	*(pred. = realized)*	*1.00%*	> 0.05 *(pass)*
Residual-hybrid (GBM)	**0.3551**	$\mathbf{-5.80}$ / $\mathbf{-5.76}$	1.19%	0.00
+ EVT tail	0.3554	-6.08 / -5.86	**1.04%**	**0.10**
GARCH(1,1)-t	0.3561	-6.17 / -5.65	1.06%	0.026
GARCH-FHS	0.3565	-6.20 / -5.59	1.05%	0.041
GJR-GARCH-skew-t	0.3574	—	—	—
EWMA (RiskMetrics)	0.3606	—	—	—
Historical simulation	0.3619	—	—	—

Diebold–Mariano, hybrid vs: GARCH-t 4.38, FHS 5.70, GJR 5.73, HS 7.34, EWMA 8.33 (all $p \approx 0$). Hybrid+EVT still beats GARCH-t (DM 3.90) and FHS (DM 4.28); its 0.0003 pinball concession to the raw hybrid (DM -5.06) buys the exception-test pass. Hybrid+EVT passes $Christoffersen_{99}$ ($p = 0.20$); no other model in the battery passes both 99% tests while topping the accuracy table. At 97.5% *every* raw model fails Kupiec; the split-conformal recalibration repairs it (GBM-recal $p = 0.24$; IQN-v2-recal $p = 0.15$, Christoffersen $p = 0.068$).

Reading guide: $Breach_{99}$ is the realized frequency of losses beyond the 99% VaR — ideal is exactly 1.00% (higher = risk understated, lower = capital wasted); the ES column compares predicted to realized average exceedance loss, ideal is equality; Kupiec p above 0.05 means the breach count is statistically consistent with 1%. An honesty note on scope: the exception statistics here pool the panel's asset-days into a single sequence, so they are *panel calibration diagnostics*, not desk-level regulatory backtests — cross-sectional exceptions on a common date are dependent, and Basel's exception counts apply to a desk's P&L time series. Per-asset Kupiec/Christoffersen pass-rate distributions (with inference clustered by date) are the registered restatement, and the "realized" ES comparator is model-dependent by construction (each model's own breaches select the conditioning set), which is why the joint Fissler–Ziegel score of the preceding paragraph, not this column, is the ranking criterion for ES. Bold marks the best value in each column. "—" entries were not recorded in the archived runs for the lower-accuracy tier (their exception statistics did not affect any ranking); completing them is scheduled with the registered battery re-run.

initially loses to trees badly (0.3650 vs 0.3551, DM 13.3) with a severely thin tail ($breach_{99}$ 3.2%); the tail-aware, monotone, conformally recalibrated version closes to a hair (0.3638 vs 0.3632, DM 3.03) with calibration repaired ($breach_{99}$ 0.92%). Trees remain the strongest tabular estimator; the tuned IQN is competitive and is the estimator that generalizes to Paper B. Third, $ES_{97.5}$ calibration — the actual capital metric — is where the hybrid's advantage is most consequential: FHS and GARCH-t both *over-state* ES by 8–11%, i.e. over-charge capital, while the hybrid is nearly exact.

The joint $(\mathrm{VaR}, \mathrm{ES})$ **score.** Because ES is not elicitable alone but the pair $(\mathrm{VaR}, \mathrm{ES})$ is jointly elicitable under the Fissler–Ziegel class (Fissler and Ziegel, 2016), the battery was re-checked under the FZ0 joint score (orientation validated by Monte Carlo against a known-truth t_5 before any ranking was read; first sandbox subsets of the panel, full-panel re-run registered). Three results. On a 45-name large-cap subset, a Gaussian-innovation GARCH is significantly *worst* at both the 2.5% and 1% levels (DM 5.4 and 6.0): under a jointly consistent score, fat tails are strictly necessary. Among the fat-tailed entrants — GARCH-t, FHS, and the hybrid+EVT

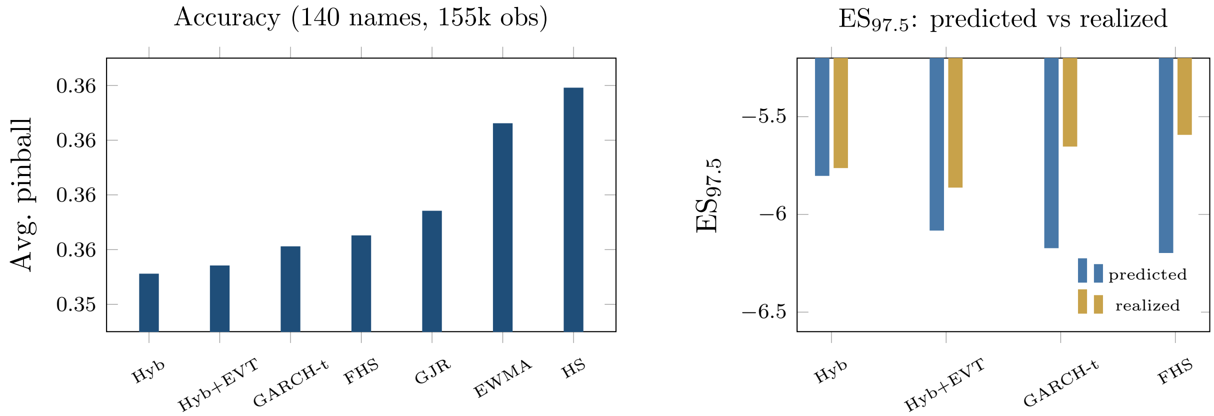

Figure 4: The FRTB battery. Left: average pinball loss by model (axis truncated to resolve the ordering; differences carry DM 4.4–8.3 vs the hybrid, all $p \approx 0$); CAViaR, not shown, statistically ties the hybrid. Right: $\mathrm{ES}_{97.5}$ predicted vs realized — FHS and GARCH-t over-state ES by 8–11% (over-charging capital) while the hybrid is nearly exact.

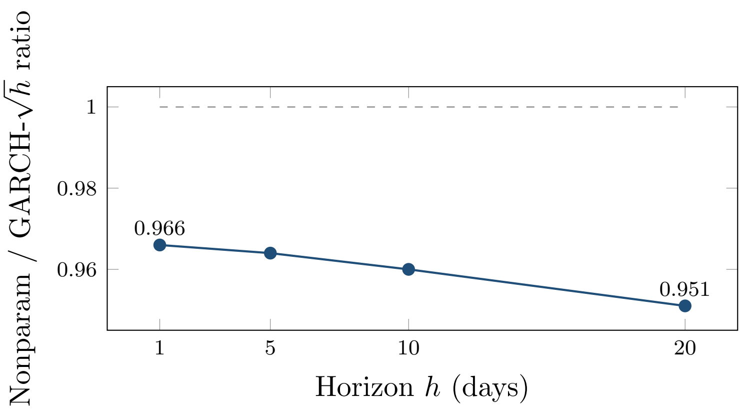

Figure 5: The edge grows with horizon. Nonparametric/GARCH pinball ratio at $h = 1, 5, 10, 20$ days (amortized panel): fat tails compound while $\sqrt{h}$ scaling compounds the shape error — ∼5% at $h = 20$.

— FZ0 differences on large caps are statistically indistinguishable ($|\mathrm{DM}| < 1.4$): on the well-behaved segment the EVT stage's distinct value is its exception-test compliance, a calibration property the FZ0 score does not reward. On a 40-name top-kurtosis subset the ordering flips and the hybrid+EVT attains the lowest FZ0 — the frontier pattern reappearing under a jointly consistent score — directionally but not yet significantly at this width (DM ≈ 0.9, $p \approx 0.17$), which is what the registered full-panel re-run is powered to decide.

5.2 The ten-day horizon and stress

At the FRTB ten-day liquidity horizon (CRSP 2014–2024, directly-learned ten-day quantiles vs GARCH $\sqrt{h}$-scaling) the pinball edge grows from ∼0.4% at $h = 1$ to ∼1.4% at $h = 10$, and holds in the 2020–2022 stress window (hybrid $1.247 <$ FHS $1.2605 <$ GARCH 1.262).

More consequentially for capital: ten-day GARCH $\sqrt{h}$-scaled ES is badly over-stated (predicted -19.4 vs realized -15.4) while the directly-learned hybrid is calibrated (-19.05 vs -17.70) with the best ten-day breach$_{97.5}$ (0.0265 vs target 0.025; GARCH 0.033). Fat tails compound with horizon; $\sqrt{h}$ Gaussian-style scaling compounds the shape error. A caveat: the temporal split places the whole test window in 2020–2024; a calm-vs-stress comparison awaits the 2000–2024 WRDS panel (registered, Section 7).

5.3 Cross-asset winners, with corrections

The clean calibrated cross-asset win is VIX (pinball 1.127 vs 1.137, DM 1.74, $p = 0.041$ one-sided, passes Christoffersen at both levels)—a marginal result: it does not survive a two-sided convention, nor a multiplicity adjustment across the 43-instrument sweep, and is reported as suggestive rather than confirmatory. Credit (52% accuracy edge, DM 30.6) and Baltic freight (15%, DM 9.6) fail breach-independence — exceptions cluster — and are reported as accuracy edges with open calibration problems, not wins. (These are the FRTB-grade single-asset fits. Freight's magnitude agrees with the lean hybrid-vs-GARCH sweep of Table 4 (ratio 0.846 there); credit does not — the lean sweep shows only 4% (DM 2.3) — because the two batteries pair different model variants against different benchmarks. Both are reported; the batteries are not interchangeable, and credit's larger figure should be read with that caveat in addition to its clustered exceptions.) An earlier electricity "win" was an artifact: log-returns on near-zero and negative power prices; refit on price *differences*, GARCH wins ERCOT (DM -2.55) and PJM (DM -2.73), and electricity is removed from the win list. The GPD tail also *hurts* some volatility series (VVIX, V2X, MOVE flip from win to tie/loss), motivating the selective-application rule of Section 3.2.

6 Applications and deployment

How a desk runs this (the whole production loop). *Nightly:* (i) update each name's GARCH filter state $\widehat{\sigma}_{t+1}$ (recursion only; parameters refit annually); (ii) assemble the state vector s_t and evaluate the shared shape model $q_\phi(\tau \mid s_t)$ — one model for the whole book, no per-name estimation; (iii) apply the frozen GPD tail below p_0 and the conformal shifts c_τ; (iv) report $\widehat{Q}_{t+1}(\tau) = \widehat{\sigma}_{t+1}\,\widetilde{q}(\tau \mid s_t)$ at the desk's levels, with ES$_{97.5}$ from the tail average; (v) read the misspecification score (9) as the model-risk dial — top decile means the parametric fallback is currently ∼2–3% wrong on that name. *Annually:* refit filters, retrain the shape model, re-estimate tail and shifts, all on trailing data. *New listing:* evaluate (ii)–(iv) from characteristics and first ticks; no history required. Total marginal cost per name per day: one filter update and one model evaluation.

Forward-looking, not just backward-looking — but a monitor, not a switch. Because the frontier score is a trailing window on GARCH-standardized residuals, it is available *ex ante*: a desk computes today's score from today's data and acts on it for tomorrow's risk number. What the evidence does *not* support is day-level model-switching. As Section 4.3

shows, the per-day oracle gap is unreachable (predicted-edge correlation +0.16), because which day a tail event lands is not forecastable from prior-day state, and a learned gate collapses to always-nonparametric. The edge is, in effect, negligible over the calm ∼90% of asset-days and concentrated (+2.7% pinball, DM 6.5) in the top misspecification decile and the most turbulent volatility regimes (an ∼fourfold gap, Section 4.3). The deployable reading is therefore not "wait, then switch" but "run the robust engine always and read the score as a dial": the nonparametric/residual-hybrid family *ties* the parametric benchmark on calm days and wins in stress, so a desk pays nothing to hold it and harvests the stress-period edge automatically. The one setting in which the score genuinely drives a switch is the per-name/standalone model, where nonparametric quantiles underperform on calm days; there the score gates, and its short detection latency (median zero, mean two to four days) costs little because the edge is back-loaded within a misspecification episode.

Regulatory capital. The hybrid+EVT+conformal engine is an internal-models VaR/ES candidate that tops the accuracy table while passing both exception tests at both levels; and because FHS/GARCH over-state $ES_{97.5}$ (by 8–11% at one day, more at ten), a calibrated engine translates directly into capital efficiency without added breach risk.

A misspecification monitor. The frontier score is a live, per-asset dial: when a desk's asset enters the top decile of residual-shape extremity, its standard VaR model is currently, locally wrong by ∼2–3% of pinball with high confidence. This is deployable as model-risk surveillance exactly because it is causal and cheap (a trailing window on GARCH residuals).

Cold-start risk for new listings. The amortized model is the only entrant in this paper that can produce a conditional risk forecast for an asset with *no return history*: an IPO, a spin-off, a newly listed coin or ETF gets day-one quantiles from its characteristics and its first ticks, because the model was trained on the pooled (state, loss) pairs of hundreds of other names. The practical numbers: the amortized forecast beats own-history benchmarks at *every* listing age with the edge largest when young (∼6–10% of pinball in the first month of trading, full-scale panel), and below roughly 250 trading days there is no fitted per-asset alternative at all — a GARCH cannot be estimated. The ablation of Section 3.3 tells a desk exactly how the machine works as history accrues: characteristics carry the forecast at birth; the name's own recent realized volatility takes over as it ages, because the model has learned one universal vol-state-to-quantile mapping rather than per-name parameters. Risk limits, margining, and market-making inventory for the newest names are the immediate use cases.

Where the edge trades — and on which side. The universes where the nonparametric tail wins — crisis FX, volatility indices, credit spreads, frontier equities — are the desks where tail-shape-aware sizing and hedging matter most. The *direction* of the trade matters as much as its location. The companion GRAFT-Q paper shows that *buying* model-flagged crash insurance loses (−1.7% per month, $t = -1.9$) even in the months the calibrated model correctly flags as elevated-tail: the crash premium survives conditioning on a correct physical-measure forecast, so a right-but-still-unprofitable buy signal is not a deployable trade. The deployable side is the other one — *warehousing* (selling) the premium — with the model supplying tail control rather than alpha on the mean: model-timed overlays improve tail outcomes, not average returns, consistent with the tail being the only place the model knows something a fixed-shape filter does not. The contribution to a premium-harvesting book is risk management (when to size down or step aside ahead of a genuine crash), not signal generation.

Scenario generation. The SBC machinery of Paper B calibrates intractable scenario-generator models (rough vol, and by extension Hawkes and jump-diffusion) with posterior uncertainty on stressed paths — a principled input to CCAR/ICAAP-style stress design.

7 Conclusion and limitations

A fixed-shape parametric filter is hard to beat exactly where its assumption holds, which is most assets on most days. The paper's answer to "when and why beat it" is a measurable frontier: local violation of the residual-shape assumption, at whose top decile the distribution-free family wins decisively, significantly, and portably across four universes. The engine that harvests this — parametric scale, nonparametric shape, EVT tail, conformal finish — meets and exceeds the FRTB standard banks actually run. The generative members of the family contribute further capabilities that no point model can — calibrated uncertainty on the risk number, and likelihood-free posteriors for models whose likelihoods do not exist in closed form — developed in the companion Paper B.

Limitations, stated plainly. Gradient-boosted trees, not the neural network, are the strongest tabular estimator; the neural IQN earns its place through sampling and amortized posterior inference, not point accuracy. The frontier score is necessary-not-sufficient (carbon/corn counterexamples; Baltic Dry wins at moderate kurtosis via limit-move dynamics). Credit and freight show large accuracy edges but clustered exceptions — unresolved. The stress analysis lacks a 2008 window pending WRDS 2000–2024 CRSP (registered). Intraday extension is blocked on tick-data access. The 99% ES credible band reaches 0.887 rather than 0.90, at $\sim$1.6$\times$ width — the honest price of deep-tail uncertainty (companion Paper B). And the per-day oracle gap of Section 4.3 should discipline the field's enthusiasm for model-switching: most of the day-level advantage of any model pair is realization noise, not predictable structure.

Registered extensions within this paper's scope, in priority order: the *full-panel* Fissler–Ziegel (VaR, ES) re-scoring (the sandbox subsets of Section 5 establish necessity of fat tails and the frontier direction; the full panel is powered for the high-kurtosis crossover) and a certainty-equivalent (MEU) translation of the ES edge into basis points of capital; the 2000–2024 WRDS panel for a 2008 stress window; and the intraday frontier once tick data is available. Companion-paper extensions (Paper B): a leverage summary statistic to identify ρ in the SBC studies, and Hawkes and jump-diffusion simulators as the next likelihood-free targets; vector quantile regression (Carlier–Chernozhukov–Galichon) as the principled multivariate transport map to succeed the scale/shape hybrid.

Data and code availability

A public replication package reproducing every table and figure in this paper — estimation code for all stages of the engine ((2)–(5)), the misspecification score (9), the FRTB battery, the generative-posterior and SBC routines, and the exact walk-forward, seeds, and hyperparameter grids referenced in the text — is archived at `github.com/ANONYMIZED/gbc-downside` with a citable snapshot at `doi.org/10.5281/zenodo.XXXXXXX` (both to be de-anonymized on acceptance). Return data derive from CRSP, Bloomberg, and OptionMetrics under the authors' institutional licenses and cannot be redistributed; the package includes the code to rebuild every panel from those sources, the derived non-proprietary series, and a synthetic example dataset (`toy_example.py`) that runs the full pipeline end-to-end without licensed data. The pre-registered tuning grid is in `TUNING_GRID_PREREG.md`; deferred analyses flagged in the text (the Fissler–Ziegel joint (VaR, ES) re-scoring, the full GPC iteration, and the 2000–2024 stress window) are tracked as open issues in the repository.

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A Proofs

The four formal results of Section 3.4 are stated with proof sketches in the body; the complete arguments, with all regularity hypotheses made explicit, are collected here. Throughout, $\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\})$ and $q_F = F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}$ is the lower generalized inverse.

Proof of Lemma 1 (pinball regret identity)

Assume $\mathbb{E}|Z| < \infty$, so

$$S(q) = (1-\tau)\int_{(-\infty,q]} (q-z)\,dF(z) \;+\; \tau\int_{(q,\infty)} (z-q)\,dF(z)$$

is finite for every q, and S is convex (an expectation of the convex maps $q \mapsto \rho_\tau(z-q)$). For fixed z, $q \mapsto \rho_\tau(z-q)$ is Lipschitz with constant $\max(\tau, 1-\tau)$, so the difference quotients $[\rho_\tau(Z-q') - \rho_\tau(Z-q)]/(q'-q)$ are bounded by the integrable constant $\max(\tau, 1-\tau)$; dominated convergence justifies differentiating under the expectation wherever the one-sided limits exist. Off the atoms of F,

$$\frac{d}{dq}\,\rho_\tau(z-q) = \mathbb{I}\{z < q\} - \tau, \qquad \Longrightarrow \qquad S'(q) = \mathbb{E}\,\mathbb{I}\{Z < q\} - \tau = F(q^-) - \tau,$$

which equals $F(q) - \tau$ for all but countably many q. Since S is convex, hence absolutely continuous, the fundamental theorem of calculus gives

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u) - \tau\big)\,du,$$

the countable atom set being Lebesgue-null; *no continuity hypothesis on F is needed for this identity*. For nonnegativity, note $q_F = \inf\{x : F(x) \geq \tau\}$ implies $F(u) \geq \tau$ for $u \geq q_F$ and $F(u) < \tau$ for $u < q_F$. If $q > q_F$ the integrand is ≥ 0 on $[q_F, q]$, so the integral is ≥ 0; if $q < q_F$,

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u) - \tau\big)\,du = \int_{q}^{q_F} \big(\tau - F(u)\big)\,du \;\geq\; 0,$$

the integrand again ≥ 0 on $[q, q_F]$. Equality holds iff $F(u) = \tau$ for Lebesgue-a.e. u between q_F and q. Finally, if F is absolutely continuous with density $0 < m \leq f \leq M$ on the interval joining q_F and q, then F is continuous there, so $F(q_F) = \tau$ and $F(u) - \tau = \int_{q_F}^{u} f$; hence $m|u - q_F| \leq |F(u) - \tau| \leq M|u - q_F|$, and integrating over the interval yields $\frac{m}{2}(q - q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2}(q - q_F)^2$. □

Proof of Proposition 1 (split-conformal validity)

Let $k = \lceil (n+1)\tau \rceil$ and assume $1 \leq k \leq n$ so that the k-th order statistic $e_{(k)} = c_\tau$ of the n calibration errors $\{e_1, \ldots, e_n\}$ exists; when $k = n + 1$ (i.e. τ within $1/(n + 1)$ of 1) set $c_\tau = +\infty$, giving coverage 1. Suppose the $n + 1$ errors $e_1, \ldots, e_{n+1}$ (calibration plus the test error $e_{n+1} = z_{n+1} - \widehat{q}(\tau \mid s_{n+1})$) are exchangeable and almost surely distinct. Then the rank R of e_{n+1} among the pooled $n + 1$ values is uniformly distributed on $\{1, \ldots, n + 1\}$. By construction $\widetilde{q}(\tau \mid s_{n+1}) = \widehat{q}(\tau \mid s_{n+1}) + c_\tau$, so

$$\{z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\} = \{e_{n+1} \leq c_\tau\} = \{e_{n+1} \leq e_{(k)}\} = \{R \leq k\},$$

the middle equality using a.s. distinctness (with ties the middle equality fails; e.g. if all e_i are equal the event has probability 1, which can exceed the stated upper bound). Therefore

$$\Pr\left(z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})\right) = \Pr(R \leq k) = \frac{k}{n+1} = \frac{\lceil (n+1)\tau \rceil}{n+1}.$$

Since $(n+1)\tau \leq \lceil (n+1)\tau \rceil < (n+1)\tau + 1$, dividing by $n + 1$ gives $\tau \leq k/(n + 1) < \tau + \frac{1}{n+1}$. The probability is over the joint law of calibration and test points; the guarantee is marginal, not conditional on s_{n+1}, because a single scalar shift c_τ pooled across states cannot deliver state-conditional coverage. □

Proof of Proposition 2 (tail wedge)

Fix $\nu > 2$, so the t_ν law has finite variance and its unit-variance version $Z = T_\nu \sqrt{(\nu - 2)/\nu}$ is well defined. The t_ν density is $\sim C_\nu |x|^{-(\nu+1)}$ as $|x| \to \infty$, so the lower tail is regularly varying with index $-\nu$: $\Pr(T_\nu < -x) = \tilde{c}_\nu x^{-\nu}(1 + o(1))$. Unit-variance scaling multiplies the constant, $\Pr(Z < -x) = c_\nu x^{-\nu}(1 + o(1))$ with $c_\nu = \tilde{c}_\nu((\nu - 2)/\nu)^{\nu/2}$, and leaves the index ν unchanged. Writing $U(\tau) = |Q_\nu(\tau)|$ and inverting the regularly varying tail (Embrechts et al., 1997) gives $\tau = c_\nu U(\tau)^{-\nu}(1 + o(1))$, hence $U(\tau) = (c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$, i.e. $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$. For $\nu_1 < \nu_2$,

$$\frac{|Q_{\nu_1}(\tau)|}{|Q_{\nu_2}(\tau)|} = \frac{(c_{\nu_1}/\tau)^{1/\nu_1}}{(c_{\nu_2}/\tau)^{1/\nu_2}}(1 + o(1)) = \text{const} \cdot \tau^{-(1/\nu_1 - 1/\nu_2)}(1 + o(1)) \longrightarrow \infty$$

because $1/\nu_1 > 1/\nu_2$. Thus for all sufficiently small τ both quantiles are negative with $|Q_{\nu_1}(\tau)| > |Q_{\nu_2}(\tau)|$, i.e. $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$. The ordering therefore holds for every $\tau < \delta$ for some $\delta > 0$; putting $\tau^* = \min(\delta, \frac{1}{2})$ gives $\tau^* \in (0, \frac{1}{2})$ as asserted. (This establishes *existence* of τ^*; the actual crossover τ_c where the ordering flips is not claimed here and is located numerically in Remark 1.) Under local truth t_{ν_1}, applying (8) with $q = Q_{\nu_2}(\tau)$ and $q_F = Q_{\nu_1}(\tau)$ bounds the regret below by $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$, with m the minimum of the t_{ν_1} density on the wedge interval. $\square$